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Nonlinear wave propagation through rigid porous materials. I: Nonlinear parametrization and numerical solutions

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It is well known that the flow resistance, as defined by Darcy's law, becomes a function of the fluid velocity when the fluid velocity becomes sufficiently large. For porous materials in air, there appears to be two separate nonlinear flow regimes: For low velocities, the resistance coefficient increases with the square of the fluid velocity; and, for high velocities, the resistance coefficient increases linearly with the magnitude of the fluid velocity. Similar regimes also exist for the mass coefficient; however, the mass coefficient is observed to decrease with increasing fluid velocity. (For pure-tone excitation, it is shown that these relationships then hold for the complex magnitude of the particle velocity.) In this paper, the two nonlinear flow regimes are parametrized for several porous materials. Nonlinear corrections are made to linear wave theory, and numerical solutions for pure tones are obtained that are then used to predict surface admittance and internal pressures of finite length sample. A simple surface pressure criterion is put fourth based upon the materials nonlinear parameters that gives an approximate maximum surface pressure that a porous material could be exposed to before nonlinear phenomena become important in describing the material's characteristics.

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INTRODUCTION

A previous article by these authors presented studies of pure-tone nonlinear wave propagation in air-saturated porous materials with some success in predicting surface phenomena (such as surface admittance and surface absorption) and internal wave attenuation.[1] In this paper, the same topic is again considered but is extended to accomplish two additional goals. The first is to give a firm foundation to the assumption in the previous article that the nonlinear effects of pure-tone fluid motion through porous materials is related to the rms particle velocity. (Specifically, in Ref. 1 it was assumed with no theoretical support that the instantaneous resistance that the fluid experiences due to the presence of the material is equal to a frequency-dependent coefficient plus a frequency-independent coefficient multiplied by the rms value of the particle velocity at that position in space.) The second is to accurately parametrize these characteristics for several porous materials of varying structure.

The apparatus from which most of the experimental data was obtained is described in Ref. 2. Since much of this paper is based upon experimental results, it is recommended that Ref. 2 be read prior to this paper in order to fully understand how the data was obtained and also to introduce the reader to some of the terminology. With these goals in mind, the paper proceeds along the following line: First, a linear theory, which describes how a fluid behaves acoustically in a porous material, is put forth; next, finite-amplitude limitations of this theory are discussed with possible corrections being suggested for a material whose flow impedance behaves nonlinearly; experimental flow impedance for three different materials are presented and tables of linear and nonlinear parameters are developed from this data; finally, finite-amplitude surface phenomena is predicted using these parameters with greater success than previously found using the more simple model presented in Ref. 1.

I. REVIEW OF THE CLASSICAL RIGID FRAME THEORY

As was done in Ref. 1, we begin this study with a review of the classical rigid frame theory, with some additional comments. We are considering sound propagation through a special form of porous material, namely, one where the pores extend through the entire sample, or are interconnected so as to allow a steady airflow through the material with only a finite resistance impeding its motion. A rigid frame implies that the porous structure (or frame) of the material does not vibrate as the fluid in the material is being acoustically excited. This is valid if the frame is very stiff or very massive. Thus we consider only a slow wave through the fluid in the pores and ignore the possible fast wave through the frame (or slow wave in case of a massive frame). Also, by assuming that wavelengths of the sound in the material are much larger than the pores, the material may be treated as a homogeneous material with bulk properties. Thus any scattering of sound in the porous frame need not be considered.

A popular model for a rigid frame material is one that considers the porous material to be comprised of many round cylinders which extend through the material and have a radius approximately equal to the pore size. With this model, the acoustic problem is reduced to solving the Navier–Stokes equations for viscous and thermally conducting fluid inside a cylinder whose walls are stationary and at a constant ambient temperature. These results are then extended to

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provide bulk properties of a porous material and has yielded good results for some materials.[3–6]

Following the notation of Wilson *et al.*,[1] the one-dimensional acoustical force and continuity equations for a bulk porous material are written as

$$-\frac{\partial P}{\partial x}=\rho_b\frac{\partial U}{\partial t}, \tag{1}$$

$$-\frac{\partial U}{\partial x}=\frac{\Omega}{K}\frac{\partial P}{\partial t}, \tag{2}$$

where P is the complex excess pressure phasor, U is the complex particle velocity phasor, Ω is the porosity (ratio of fluid in a given volume to the total volume considered), and ρ_b and K are complex, frequency-dependent coefficients for the density and bulk modulus, respectively. {Note that throughout this paper P and U will represent their peak values, not rms. Also, a phasor will be defined here as a complex, spacially dependent amplitude function multiplied by an exponential time dependence [e.g., $P=A(x)\exp(j\omega t)$, where $A(x)$ is some complex function of x].}

Interpreting the density of a viscous fluid as a complex number is convient and easy to verify if we examine the physical meaning of its real and imaginary parts. Writing the complex density as $\rho_b(\omega)=M(\omega)+R(\omega)/j\omega$, where M and R are real functions, and using $\partial/\partial t=j\omega$ in Eq. (1), it quickly becomes apparent that M is a mass reactance term and R is a resistive damping term due to the viscous drag of the fluid passing through the porous material. On the other hand, the necessity of having a complex bulk modulus becomes apparent when one considers the thermal interaction of the frame and the fluid. Since, for most materials, the heat capacity of the material will be much greater than that of the fluid (especially air), the frame will be kept as a constant ambient temperature. Thus when a volume of the fluid experiences temperature variations due to the acoustic compressions and expansions, the temperature of the volume will be out of phase with the pressure due to the thermal conduction of heat between the fluid and the frame. This phase is then incorporated into the bulk modulus, making it a complex value. The magnitude of K in air will also change as it goes from a nearly isothermal value P_0 for very low frequencies to a nearly adiabatic value γP_0 for very high frequencies where P_0 and γ are the static pressure and the ratio of specific heats of air, respectively.

Functional form of ρ_b and K for a cylindrical pore geometry are[3,4,6]

$$\rho_b=q^2\rho_0\Omega/[1-S(\kappa\sqrt{-j})], \tag{3}$$

$$K=\gamma P_0/[1+(\gamma-1)S(\kappa\sqrt{-j\gamma N_{\rm pr}})], \tag{4}$$

where q is called the tortuosity (representing the slanting of the pores), ρ_0 is the static density of fluid in the pores, and S is a function defined by

$$S(z)=2J_1(z)/zJ_0(z), \tag{5}$$

where J_1 and J_0 are the first- and zeroth-order Bessel functions of the first kind, respectively, $N_{\rm pr}$ is the Prandtl number, and κ is a geometric-frequency scaling parameter defined by

$$\kappa=a_e\sqrt{\omega/\nu}. \tag{6}$$

In Eq. (6), a_e is an effective pore size, ω is the radian frequency, and ν is the kinematic viscosity of the fluid. Usually the effective pore size a_e may be taken to be approximately equal to twice the hydraulic radius defined as the ratio of fluid in a volume divided by the wetted surface area in that volume.[7] However, this value is usually multiplied by a value (frequently called the shape factor) within the range 0.817–1.0 (see Ref. 6) to obtain a good fit to experimental data.[5] Finally, a useful equation relating the ratio q/a_e to the static flow resistance[1,8,9] σ_0 may be obtained[1,6] as

$$\sigma_0=8(\rho_0\nu/\Omega)(q/a_e)^2. \tag{7}$$

Since σ_0 is a parameter that may be measured nonacoustically, Eq. (7) may be used to define a value for q and thus we are left with only choosing the exact value of a_e by acoustical methods.

The rigid frame theory of Eqs. (1)–(7) has had much success in predicting surface admittance[3,5] and internal wave propagation[4,6] for those materials whose frame seems to behave rigidly in the range of frequencies investigated. Thus we choose to work with this relatively simple theory (as opposed to a nonrigid frame theory which must also deal with frame motion and inherent fluid/frame wave coupling)[3,10,11] in trying to understand nonlinear effects that become dominant at high sound-pressure levels.

II. NONLINEAR CORRECTIONS TO COMPLEX DENSITY AND BULK MODULUS

Equations (1) and (2) form a linear theory for porous materials in as much as they are a set of linear differential equations in x when the $\partial/\partial t$ operator is replaced by $j\omega$. However, it has been well established that certain porous materials will behave nonlinearly when excited by high intensity sound. In a recent article in this *Journal*, Kuntz and Blackstock[9] show data of pressure attenuation in long (considered to be infinite) porous samples under excitation by a single high intensity tone. Their data clearly show that for the materials studied (foams and batted Kevlar) and at high SPLs (up to 170 dB), wave attenuation in the materials is not purely exponential as is predicted by all linear theories. Thus we wish to find corrections to the linear theory of Eqs. (1) and (2), which will include the nonlinear properties of the fluid in the material and properly predict its behavior.

A. Nonlinear viscous effects

A commonly measured parameter of open-cell porous materials is its flow resistance σ. From Darcy's law, the flow resistance is defined to be the pressure drop per unit length divided by the particle velocity passing through the material,[8] or

$$\sigma=\frac{-dp_{dc}/dx}{u_{dc}}, \tag{8}$$

where the subscript dc denotes a steady, nontime-varying fluid flow. The flow resistance is caused by the viscous drag the fluid experiences as it passes through the pores. It would seem reasonable that σ should be a constant, since one would expect that if the fluid velocity (u_{dc}) doubles, the pressure drop across the material would also double. However, it has been well established that viscous drag is generally a nonlinear phenomena. Even for the simple case of fluid passing by a cylinder, the drag is not linearly related to the velocity of the fluid.[12] Thus, since σ is due to the viscous drag, we cannot expect σ to be a constant, independent of p_{dc} and u_{dc} in Eq. (8). This nonlinear behavior of σ has been known for some time. In 1901, Forchheimer[13] suggested that σ was not a constant, but rather was a function of u_{dc}. If one assumes that the velocity dependence of σ can be described by a power series of the particle velocity, then one may write

$$\sigma = \sigma_0 + \eta_0 u_{dc} + \eta_1 u_{dc}^2 + \eta_2 u_{dc}^3 \cdots, \tag{9}$$

where σ_0 is a linear flow resistance valid for low fluid velocities, and η_n will be called the Forchheimer coefficients. Note that σ_0 is the value usually reported as the flow resistance of a porous material. Since Eq. (9) involves an infinite number of coefficients, it has been a common practice to employ only the first nonlinear term as a correction to Darcy's law.[6,7,9] Thus σ is commonly approximated by $\sigma \cong \sigma_0 + \eta_0 u_{dc}$, when nonlinear drag effects are believed to be important. However, it will be shown here that this is not always a good form for acoustic applications.

As previously argued by these authors,[1] since σ is just the zero frequency limit of $j\omega\rho_b$ [as can be seen from Eq. (1)], then ρ_b itself must be a function of the particle velocity. But, how to choose the velocity dependence of ρ_b is not clear. Previously,[1] the velocity dependence took the form of adding an $\eta_0 |U|/(j\omega\sqrt{2})$ term to the linear complex density given in Eq. (3), where $|U|$ is the magnitude of the complex particle velocity. But why does it take this form? (Note that in Ref. 1, $|U|$ was defined as the rms particle velocity. However, in this paper, $|U|$ will be defined as the magnitude of the peak complex particle velocity U.) The reason for the nonlinear correction to the complex density becomes partially clear when ρ_b is written in the form $M + R/j\omega$. If we denote the velocity corrected nonlinear complex density by ρ_{bnl}, then with the $\eta_0|U|/(j\omega\sqrt{2})$ correction it becomes

$$\rho_{bnl}(\omega) = M(\omega) + \left[R(\omega) + \eta_0 |U|/\sqrt{2}\right]/j\omega, \tag{10}$$

where the frequency dependence has been explicitly shown. As can be seen from this equation, the correction term is just a first term Forchheimer correction of Eq. (9) where the rms particle velocity has replaced the steady flow velocity u_{dc}. The reason for only using a first term correction is self-evident. However, using $|U|/\sqrt{2}$ in place of u_{dc} is not, and it was supported mainly by the fact that it gave good theoretical predictions. Of course, this reasoning is not satisfactory if one wishes to properly understand the physics of the nonlinear acoustic behavior.

To better understand why Eq. (10) gave good results, we begin by rewriting the force equation of Eq. (1) in the quasiinstantaneous form:[1,5,14]

$$-\frac{\partial p}{\partial x} = R(\omega,u)u + M(\omega,u)\frac{\partial u}{\partial t}, \tag{11}$$

where p and u are now the instantaneous acoustic pressure and particle velocity, respectively, and where we have assumed that the resistive and mass coefficients now have an instantaneous velocity dependence. Zorumski and Parrott[14] have measured $R(\omega,u)$ and $M(\omega,u)$ for porous materials resembling thin perforates and found R to increase with increasing u and to be roughly frequency independent, while M decreased with increasing u and had a definite frequency dependence.

[Note that, since the coefficients of Eq. (11) have a frequency dependence, Eq. (11) is not a true differential equation but does provide a convenient form to work with and to visualize the physical problem. Also, since the equation is nonlinear, harmonics will be generated that makes choosing unique forms for R and M a nontrivial problem (see Ref. 5 Section 2.2). In the rest of the paper we will assume that both u and p are dominated by a single pure tones of fundamental radian frequency ω and that any higher harmonics may be ignored.]

Now, we next assume that both $R(\omega,u)$ and $M(\omega,u)$ may be expanded in a power series in terms of u as

$$R(\omega,u) = r_0(\omega) + r_1(\omega)u + r_2(\omega)u^2 + r_3(\omega)u^3 + \cdots, \tag{12a}$$

$$M(\omega,u) = m_0(\omega) + m_1(\omega)u + m_2(\omega)u^2 + m_3(\omega)u^3 + \cdots, \tag{12b}$$

and that the particle velocity can be approximated by a single pure tone as

$$u(x,t) \cong |U(x)|\cos[\omega t + \theta(x)]. \tag{13}$$

[Note that since U was defined to be the complex particle velocity phasor, which has spacial dependence, we need to take its magnitude in order to use it in Eq. (13).] Substituting Eqs. (12) and (13) into Eq. (11) and balancing the harmonics, we find for the fundamental[5]

$$\frac{\partial p}{\partial x} = \left(r_0(\omega) + \frac{3}{4}r_2(\omega)|U|^2 + \frac{5}{8}r_4(\omega)|U|^4 + \cdots\right)u + \left(m_0(\omega) + \frac{1}{4}m_2(\omega)|U|^2 + \frac{1}{8}m_4(\omega)|U|^4 + \cdots\right)\frac{\partial u}{\partial t}. \tag{14}$$

This quasiinstantaneous form may be rewritten in the phasor form[5]

$$\frac{\partial P}{\partial x} = \rho_{bnl}(\omega,|U|)\frac{\partial U}{\partial t}, \tag{15}$$

where $\rho_{bnl}(\omega,|U|)$ is a new velocity-dependent nonlinear complex density that can be written as

$$\rho_{bnl}(\omega,|U|) = \rho_b(\omega) + \tfrac{1}{4}m_2(\omega)|U|^2 + \tfrac{1}{8}m_4(\omega)|U|^4 + \cdots + \frac{1}{j\omega}\left(\frac{3}{4}r_2(\omega)|U|^2 + \frac{5}{8}r_4(\omega)|U|^4 + \cdots\right) \tag{16}$$

and where $m_0(\omega) + r_0(\omega)/j\omega$ is just the linear complex density, $\rho_b(\omega)$, of Eq. (3). [Again, note that $|U|$ of Eq. (16) is the magnitude of the complex particle velocity which, here, is taken to be the peak value and not an rms value.]

Comparing the nonlinear complex density of Eq. (16), which has been obtained by harmonically balancing the nonlinear instantaneous force equation, to ρ_{bnl} of Eq. (10), which was obtained semiempirically, we find several things worth noting. First, the origin of the dependence of ρ_{bnl} on the magnitude of the particle velocity $|U|$ has been clearly established and thus at least partially explains why Eq. (10) gave results that agreed with experiment. Also, note that ρ_{bnl} of Eq. (16) does not depend upon any of the odd expansion terms of R and M shown in Eq. (12). This has two significant implications.

The first is that the semiempirical form for ρ_{bnl} of Eq. (10) used by Wilson *et al.*[1] must be incorrect, at least in the strictest sense. Second, since σ of Eq. (8) is the zero frequency limit of $j\omega\rho_{bnl}$, we must have that $\eta_n = 0$, where n is odd. However, this result is not surprising. If one assumes that the material is homogeneous and isotropic, then the flow resistance would be independent of the direction of flow and thus

$$\sigma(u_{dc}) = \sigma(-u_{dc}). \tag{17}$$

Comparing Eqs. (17) and (9) yields the result that $\eta_n = 0$ for n odd. Thus the result obtained from harmonic balance is consistent with physical reasoning.

At this point, one is interested in the actual form that the nonlinear velocity dependence of ρ_{bnl} in Eq. (16) takes on for real materials. However, since analytical first principle models for this behavior are not currently available, we were forced to measure ρ_{bnl} for all materials of interest. As discussed in a companion paper,[2] it is possible to design an experimental apparatus that measures an approximation to the dynamic flow impedance Z_f defined by

$$Z_f \equiv \frac{-dP/dx}{U}. \tag{18}$$

The measured approximation of Z_f will be denoted by Z_{fm}. How Z_{fm} is measured and how it is related to Z_f is discussed in detail in Ref. 2. [Note that in Eq. (18), P and U are phasors, and hence Z_f will in general be complex and frequency dependent.] A comparison of Eq. (18) to the nonlinear force Eq. (15) shows that

$$Z_f = j\omega\rho_{bnl}(\omega,|U|). \tag{19}$$

Thus there is a direct and simple relation between ρ_{bnl} and Z_f.

By comparing Eq. (19) with Eq. (16), the physical significance of the real and imaginary parts of Z_f become apparent. In doing so, we define new pure-tone resistive and mass coefficients as

$$R(\omega,|U|) = \mathrm{Re}[Z_f] = r_0(\omega) + \tfrac{3}{4}r_2(\omega)|U|^2 + \tfrac{5}{8}r_4(\omega)|U|^4 + \cdots, \tag{20a}$$

$$M(\omega,|U|) = \mathrm{Im}[Z_f/\omega] = m_0(\omega) + \tfrac{1}{4}m_2(\omega)|U|^2 + \tfrac{1}{8}m_4(\omega)|U|^4 + \cdots, \tag{20b}$$

where the $(\omega,|U|)$ dependence shows that these functions are harmonically averaged quantities and do not represent the instantaneous quantities of Eq. (12). [Unless specifically denoted by a (ω,u) dependence, throughout the rest of the paper the symbols R and M will refer to the quantities of Eq. (20) and not to Eq. (12), as we will become solely interested in pure tones.] Thus the real part of Z_f corresponds to a resistive effect and the imaginary part corresponds to a mass loading effect. Combining Eqs. (19) and (20), one quickly sees that

$$Z_f = R(\omega,|U|) + j\omega M(\omega,|U|). \tag{21}$$

Shown in Figs. 1–3 are measured $R_m(\omega,|U|)$ and $M_m(\omega,|U|)$ characteristics for a variety of materials (where $Z_{fm} = R_m + j\omega M_m$ again refers to the "measured" flow impedance). We studied: open cell porous foams (previously studied in Refs. 15 and 10), uniform nonconsolidated glass beads (studied in Refs. 5 and 6), and layered fine-fibered Kevlar (studied in Refs. 7 and 11). [Note that in the dynamic flow impedance apparatus that was used to measure Z_{fm} (discussed in Ref. 2), the samples lengths for the foams were 5 cm in length while the glass bead and Kevlar samples were 1 cm due to their much higher flow impedance.]

From these figures it can be seen that the formulation of Eq. (20a), where $R(\omega,|U|)$ varies according to $r_0 + \frac{3}{4}r_2|U|^2$ for sufficiently small velocities appears to be valid. This can be seen by noticing that in the plots, near the origin

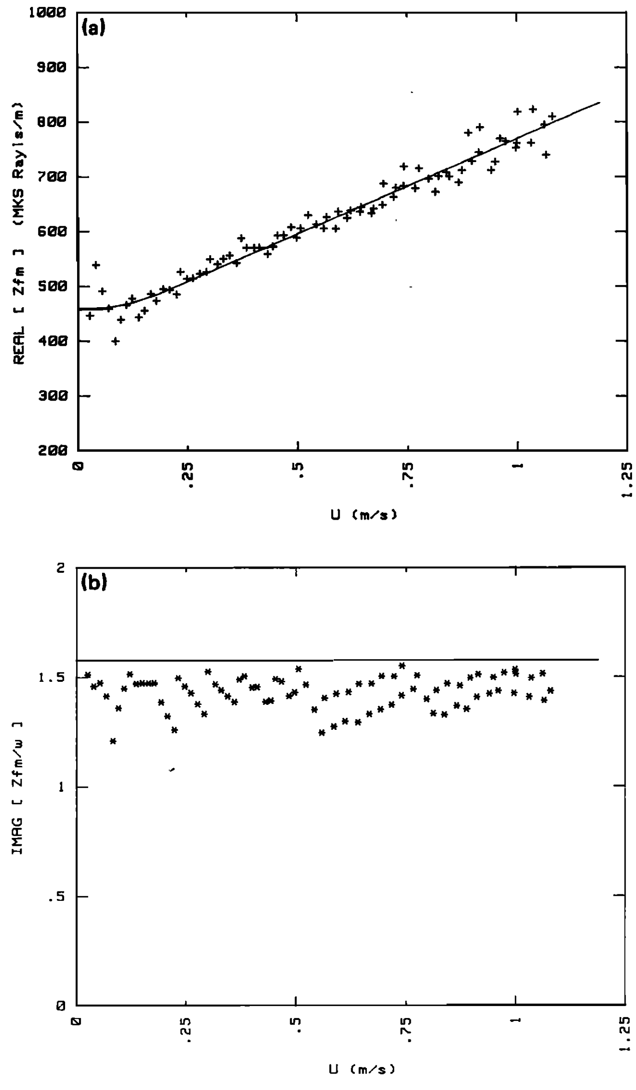

FIG. 1. Measured flow impedance Z_{fm} vs peak particle velocity for 30-ppi foam taken at a frequency of 182 Hz on a 5 cm sample. Nonlinear parameters used to create solid curves are: $U_c = 0.2$ m/s and $r_1 = 410$ mks Rayls s/m^2. Linear parameters are: $\Omega = 0.973$, $a_c = 0.738$ mm, and $\sigma_0 = 295$ mks Rayls/m. (a) $\mathrm{Re}[Z_{fm}]$ and (b) $\mathrm{Im}[Z_{fm}/\omega]$ vs $|U|$.

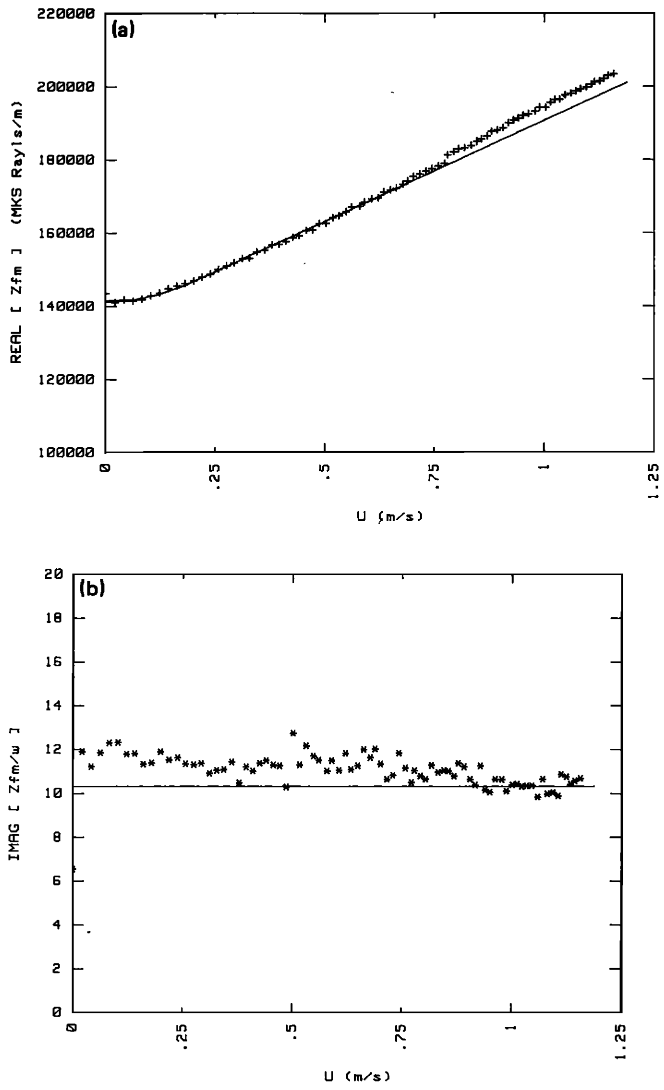

FIG. 2. Measured flow impedance Z_{fm} vs peak particle velocity for 0.5-mm glass beads taken at a frequency of 389 Hz on a 1-cm sample. Nonlinear parameters used to create solid curves are: $U_c = 0.2$ m/s and $r_1 = 65\,000$ mks Rayls s/m^2. Linear parameters are: $\Omega = 0.392$, $a_c = 0.0823$ mm, and $\sigma_0 = 141{,}100$ mks Rayls/m. (a) shows $\mathrm{Re}[Z_{fm}]$ and (b) shows $\mathrm{Im}[Z_{fm}/\omega]$ vs $|U|$.

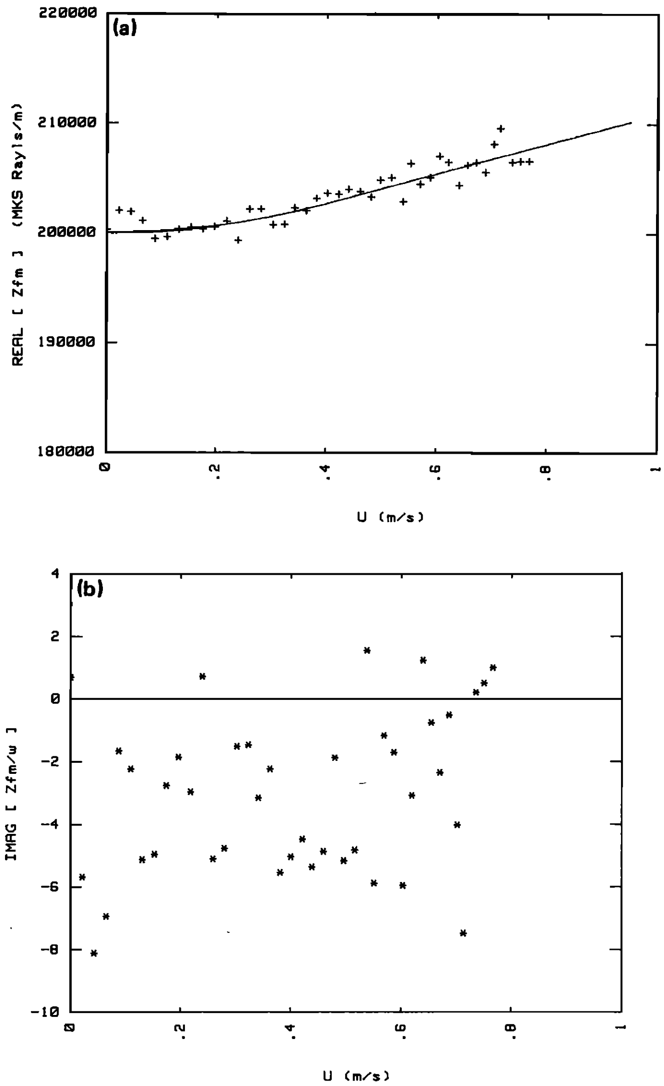

FIG. 3. Measured flow impedance Z_{fm}, vs peak particle velocity for layered Kevlar taken at a frequency of 166 Hz on a 1-cm sample. Nonlinear parameters used to create solid curves are: $U_c = 0.4$ m/s and $r_1 = 16\,000$ mks Rayls s/m^2. Since Kevlar behaves as a soft frame for low frequencies, the mass term is negative showing that the fluid is behaving "springlike" rather than "masslike," as is predicted by the rigid frame theory. Thus the linear rigid frame theory is not applicable to Kevlar at this low of frequency. (a) shows $\mathrm{Re}[Z_{fm}]$ and (b) shows $\mathrm{Im}[Z_{fm}/\omega]$ vs $|U|$.

$(|U| = 0)$, the resistive term increases along a curve and has a zero slope at the origin. This can be approximated by some degree of accuracy by a quadratic fit as given above, from which we also conclude that the even term expansions for pure tones are valid. However, as the fluid velocity increases, a transition occurs and the dependence of R upon $|U|$ appears to become linear or $R(\omega,|U|) = r_0' + (8/3\pi)r_1|U|$ where the $8/3\pi$ term comes from a harmonic balancing operation of the instantaneous form $R(\omega,u) = r_0' + r_1 u$. [Note that this is the same operation that yielded the 3/4, 5/8, etc., coefficients of Eq. (14).] This linear appearance of the velocity dependence is not explicitly predicted by the even power series expansion of Eq. (20a) but neither is it explicitly disallowed. For support of this statement we point out that if the coefficients of an infinite even powered expansion, such as in Eq. (20), are chosen properly, any function symmetrical about the origin may be obtained. Even if the function is $f(x) = |x|$, x real. Thus a linear dependance of $R(\omega,|U|)$ upon $|U|$ is possible if the r_n coefficients of Eq. (20a) are chosen properly. However, this rationalization does not explain the physics behind the two apparent regions of flow resistance: quadratic for low velocities, and linear for high velocities.

While $R(\omega,|U|)$ shows definite nonlinear behavior, the mass term $M(\omega,|U|)$ shows little variation. In Fig. 1(b) it appears to be constant, while in Fig. 2(b) it is seen to decrease slightly with increasing particle velocity. That is, the change in the mass term with $|U|$ is small compared to its magnitude. Finally, the Kevlar data in Fig. 3(b) show a very scattered mass term behavior which is generally negative, implying that the fluid in Kevlar behaves springlike rather that masslike. This is believed to be due to a flexible (rather that rigid) frame.[3,10,11] However, for the other materials it was generally observed that the mass term did decrease with increasing particle velocity as was also observed by Zorumski and Parrott.[14] From the data taken, it is seen that for our range of velocities, M decreases roughly linearly with increasing particle velocity, or $M(\omega,|U|) \propto m_1(\omega)|U|$ where $m_1(\omega)$ is a frequency-dependent nonlinear coefficient similar to r_1. But we know from Eq. (20b) that near $|U| = 0$, $M(\omega,|U|)$ must vary quadratically, or $M(\omega,|U|) \cong m_0(\omega) + (1/4)m_1(\omega)|U|^2$ for small $|U|$. Thus our proposed final analytical form for $M(\omega,|U|)$ takes on a similar form as that for $R(\omega,|U|)$.

Under these approximations, the final simplified form based upon theoretical and experimental results for $R(\omega,|U|)$ and $M(\omega,|U|)$ are

$$R(\omega,|U|) = \begin{cases} r_0(\omega) + \frac{3}{4}r_2(\omega)|U|^2, & |U| \leqslant U_c, \\ r_0'(\omega) + (8/3\pi)r_1(\omega)|U|, & |U| > U_c, \end{cases} \tag{22a}$$

$$M(\omega,|U|) = \begin{cases} m_0(\omega) + \frac{1}{4}m_2(\omega)|U|^2, & |U| \leqslant U_c, \\ m_0'(\omega) + (4/3\pi)m_1(\omega)|U|, & |U| > U_c, \end{cases} \tag{22b}$$

where $r_0(\omega)$ is the linear, low velocity limit of $R(\omega,|U|)$ and is easily shown to be equal to $\mathrm{Re}[j\omega\rho_b(\omega)]$ as seen in Eq. (3); $r_2(\omega)$ is the frequency-dependent first term even expansion coefficient of $R(\omega,|U|)$; and $r_1(\omega)$ is the "linear" nonlinear resistive coefficient chosen so that the linear slope of $R(\omega,|U|)$ for high fluid velocities is $(8/3\pi)\, r_1(\omega)$. The parameter U_c is a critical (or transition) velocity where the dependence of $R(\omega,|U|)$ upon $|U|$ goes from quadratic to linear and is chosen so that $\partial R(\omega,|U|)/\partial|U|$ is piecewise continuous, and $r_0'(\omega)$ is an offset value from $r_0(\omega)$ and is chosen so that $R(\omega,|U|)$ is continuous. Similar comments apply for the $m_0(\omega)$, $m_0'(\omega)$, $m_1(\omega)$, and $m_2(\omega)$ coefficients. Note that the same critical velocity was used in describing the transition from quadratic to linear regimes for both R and M. This was done because the observed variation in M over the velocities attained were too small to allow a distinct U_c to be evaluated. Thus U_c is determined from $R(\omega,|U|)$ data alone. Whether the mass term would have a critical velocity much different from the resistive term is not known, but it would seem reasonable that they would be similar, if not identical.

Under these restrictions, it can be shown that

$$r_2(\omega) = (16/9\pi)[r_1(\omega)/U_c], \tag{23a}$$

$$m_2(\omega) = (8/3\pi)[m_1(\omega)/U_c], \tag{23b}$$

$$r_0'(\omega) = r_0(\omega) - (4/3\pi)r_1(\omega)U_c, \tag{23c}$$

and

$$m_0'(\omega) = m_0(\omega) - (2/3\pi)m_1(\omega)U_c. \tag{23d}$$

B. Nonlinear thermal effect

From the standpoint of the linear theory of Eqs. (1) and (2), there are two possible ways to account for nonlinearities. They are, of course, through the complex density coefficient ρ_b and the complex bulk modulus coefficient K. Previously in this section we have seen that viscous nonlinearities are responsible for the departures in the ρ_b coefficient which causes it to become dependent upon the magnitude of the particle velocity for pure-tone excitations. Since K is mainly a thermodynamically controlled parameter, one may expect that it would become pressure dependent, since the pressure and temperature of the fluid are related in these processes. However, the way in which any temperature effects would be first observed is through the parameters ν, γ, and $N_{\rm pr}$ which are the kinematic viscosity, ratio of specific heats, and the Prandtl number, respectively [see Eqs. (3), (4), and (16)]. Although these quantities do have some temperature dependence, it is not very great. Of the three parameters, $N_{\rm pr}$ has the greatest temperature dependence with a $\pm 1\%$ deviation from its room temperature value occurring at -3.1 °C ($+1\%$) and 54 °C (-1%). [The ratio of specific heats is the next most sensitive with its $\pm 1\%$ deviations occurring at -153 °C ($+1\%$) and 216 °C (-1%).] If one assumes a strict adiabatic thermal process and γ a constant of 1.4, then it can be shown that a 1% change in $N_{\rm pr}$ corresponds to an rms acoustic pressure of 183 dB. Even this small change occurs at a pressure currently beyond our interest and well beyond our experimental capabilities. Also, for low frequencies, the thermal process is isothermal and so no nonlinear thermal departure would be possible. Therefore, one is led to conclude that, in general, K will experience no significant dependence upon the dynamic pressure P.

This prediction has been verified experimentally in the same apparatus from which Z_{fm} measurements were taken. The experiments are analogous to the Z_{fm} experiments described in Ref. 2. It is possible to arrange the experiment such that there is a pressure maximum within the mounted sample so that a large velocity gradient appears across it. In such a situation, it is possible to define a "measured" complex bulk modulus, based upon Eq. (2), as

$$K_m = \frac{j\omega\Omega P_0}{-(U_2 - U_1)/d}, \tag{24}$$

where U_1 and U_2 are the particle velocities on either side of the sample, d is the sample thickness, and P_0 is the average pressure in the sample. [See Sec. I of Ref. 2 for a full description of a "measured" value of Z_f. A discussion of Eq. (24) would be similar.] This quantity K_m has been measured for the materials studied in this paper for pressure up to 160 dB and found to have no noticeable change. Thus we conclude that thermal nonlinearities are not significant in porous materials.

III. PARAMETRIZATION OF NONLINEARITIES OF POROUS MATERIALS

In using the rigid frame model presented in Sec. I, the three parameters required to characterize the linear behavior of rigid frame porous materials are a_e, Ω, and σ_0. In the

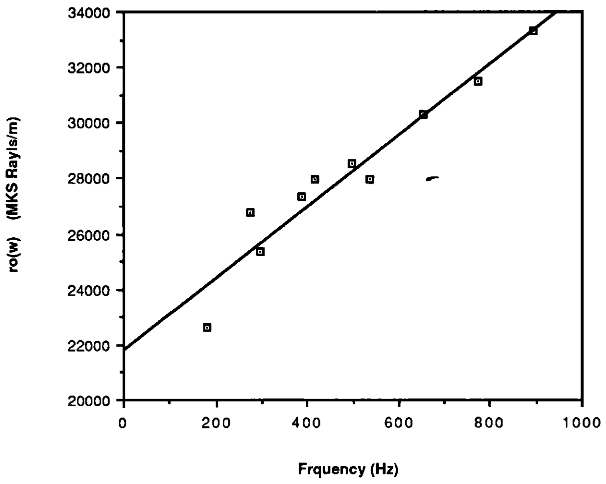

FIG. 4. Measured linear flow resistance term $r_0(\omega)$ vs frequency for 1-mm glass beads taken from Z_{fm} measurements. The curve is a linear regression fit of the data given by $r_0(\omega) = 12.9f + 21\,800$ (correlation $= 0.96$), where f is the frequency in Hz.

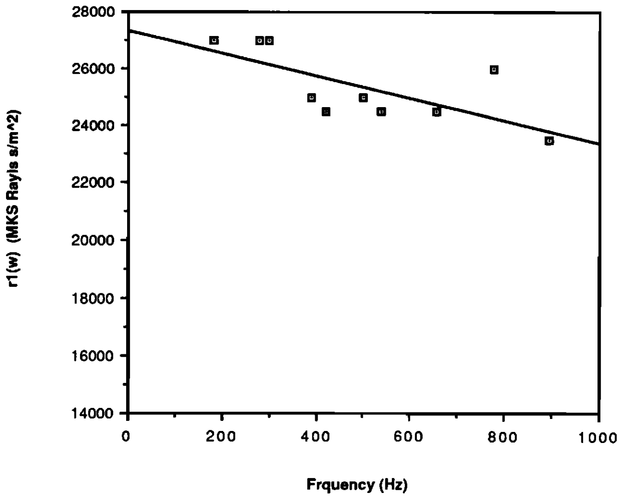

FIG. 6. Measured nonlinear resistive coefficient r_1 vs frequency for 1-mm glass beads taken from Z_{fm} measurements. The curve is a linear regression fit of the data given by $r_1(\omega) = -3.91f + 27\,300$ (correlation $= 0.70$), where f is the frequency in Hz.

previous section, it was shown that the nonlinear viscous behavior of these materials may be described by the three additional parameters U_c, r_1, and m_1. Of these parameters, all but the volume porosity Ω can be obtained from a series of Z_{fm} measurements such as those shown in Figs. 1–3.

Shown in Figs. 4–8 are material parameters for 1-mm-diam beads obtained from Z_{fm} plots. As discussed in Ref. 2, σ_0 is the zero frequency limit of R_{om} (where again the subscript m refers to the fact that this is a measured quantity of a finite-length sample);[2] thus from Fig. 4, a value of 21 800 mks Rayls/m is obtained for 1-mm beads.

Now, by observing from Fig. 5 that $M_{om}(\omega) \cong 9.4$, we obtain from Eq. (32) of Ref. 2 that $M(0) = 9.45$, where a sample length of 1 cm was used. Then, applying Eq. (31) of Ref. (2), a value of 0.194 mm is obtained for a_e.

Shown in Fig. 6 is a plot of r_1 vs frequency. Here, r_1 shows a slight frequency dependence whose trend is for r_1 to decrease with increasing frequency. This dependence was observed to be even greater in the foams. Figure 7 shows the frequency dependence of the m_1 coefficient. As with the other materials tested, m_1 is seen to decrease with increasing frequency. Unfortunately, due to the limited range of fre-

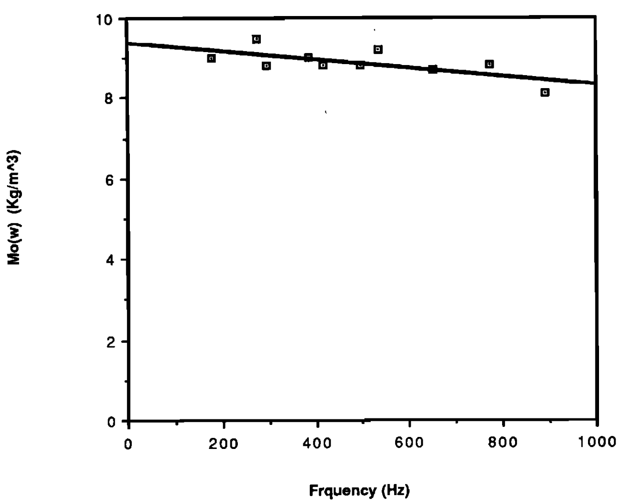

FIG. 5. Measured linear mass term $M_{om}(\omega)$ vs frequency for 1-mm glass beads taken from Z_{fm} measurements. The curve is a linear regression fit of the data given by $M_{om}(\omega) = -0.001f + 9.40$ (correlation $= 0.69$), where f is the frequency in Hz.

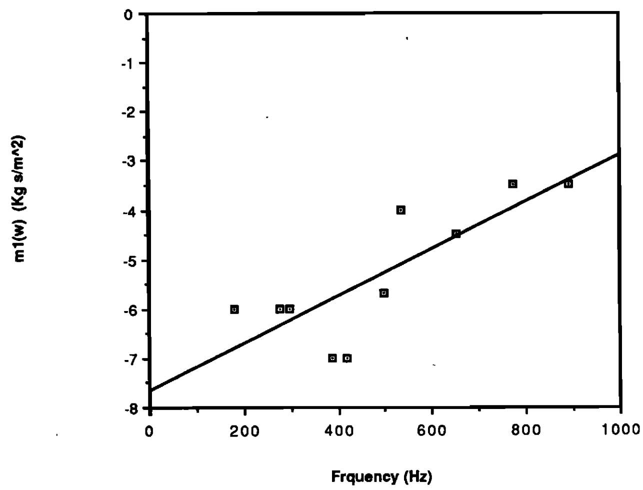

FIG. 7. Measured nonlinear mass coefficient m_1 vs frequency for 1-mm glass beads taken from Z_{fm} measurements. The curve is a linear regression fit of the data given by $m_1(\omega) = 0.0048f - 7.66$ (correlation $= 0.81$), where f is the frequency in Hz.

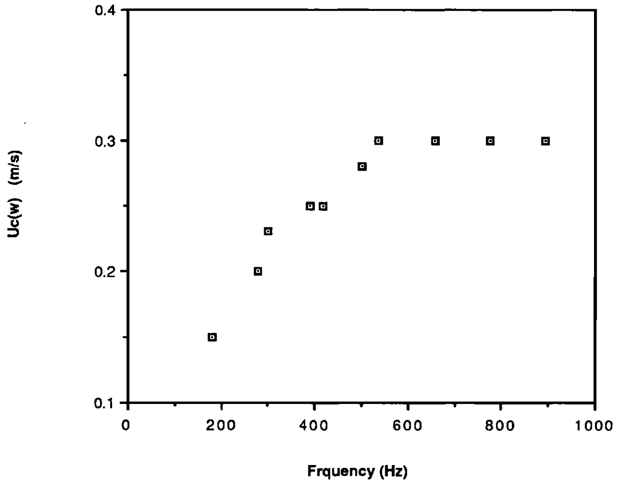

FIG. 8. Measured values of the critical velocity U_c vs frequency for 1-mm glass beads taken from Z_{fm} measurements.

quency measurements, the limits r_1 and m_1 for large frequencies cannot be established by our data. However, from high intensity admittance measurements it is believed that m_1 will go to zero. The r_1 coefficient appears to continue to decrease for increasing frequencies; however, the rate of decrease also decreases and it is believed that before r_1 becomes zero, the wavelengths in the porous material would approach the average pore size thereby making the Rayleigh tube model invalid.

Finally, Fig. 8 shows experimental values of U_c obtained for the same range of frequencies. The plot shows a range (below 500 Hz) where U_c is increasing with frequency. Above 500 Hz, it appears to level off and become constant. However, this behavior is atypical of the materials studied. The other materials studied (i.e., 0.5-mm beads, 100- and 30-ppi foams) showed U_c to be scattered about some mean value with no clear frequency dependence. Therefore, we choose to take U_c to be a frequency-independent parameter and from the data of Fig. 8, fix it to be 0.3 m/s for the 1-mm beads.

From similar data as that shown in Figs. 4–8, the parameters of Tables I and II were compiled for the other porous materials. Note that Table I gives the values of a_e and σ_0 computed using Z_{fm} data and also those values found by chosing a_e and σ_0 which best fit admittance data taken in a standing wave tube.[5] Since, in our work, we are usually interested in surface admittance of finite length samples, the a_e and σ_0 values obtained by fitting admittance data are taken to be more accurate than values based on the flow impedance approach. We observe that the flow impedance approach for obtaining these values typically yielded results that are larger than expected. It is not known why such uniform errors occur and the fractional error is not constant. However, if one ignores the 0.5-mm bead results, then the values computed using Z_{fm} measurements are within 12% of those obtained from admittance measurements. This appears to be an encouraging result and lends confidence to the method.

IV. HIGH-INTENSITY SURFACE ADMITTANCE AND PRESSURE FIELD MEASUREMENTS

In this section, we concern ourselves with predicting the acoustic behavior of a porous material under finite-amplitude sound excitation. Specifically, data of the surface admittance of finite-length samples versus surface pressure at fixed frequencies are presented. The experimental data are compared to theoretical results predicted using the parameters of Tables I and II and the numerical solution technique described in the Appendix of Ref. 1.

The surface admittance of the samples were taken using both the standing wave apparatus described in Ref. 1 and the apparatus of Ref. 2 The samples were under "pure-tone" (or fixed frequency) excitation. Harmonics were observed to be at least 10 dB lower that the fundamental. The surface admittance presented is normalized with respect to $\rho_0 c$; thus the admittance y is

$$\begin{aligned} y &= \rho_0 c[U/P]|_{\text{surface}} \\ &= g + jb, \end{aligned} \tag{25}$$

where U and P are velocity and pressure phasors, g is the normalized conductance, and b is the normalized susceptance. The samples are of finite length backed by a rigid termination which yields the boundary condition that $U = 0$ at the termination.

Finite-length samples were used for two reasons. First, samples of infinite extent (long enough so that an internally reflected wave from the termination could be ignored) yielded poor results. This was believed to come from sample inho-

TABLE I. A compiled list of the linear parameter values for the five porous materials studied. An x for Kevlar indicates that the value was not computable.

Material	From Z_{fm} measurements		From admittance measurements		
	a_e (mm)	σ_0 (Rayls/m)	a_e (mm)	σ_0 (Rayls/m)	Ω
30-ppi foam	0.828	295	0.738	295	0.973
100-ppi foam	0.185	5960	0.166	5470	0.974
0.5-mm beads	0.105	127 000	0.0823	98 100	0.392
1-mm beads	0.194	21 800	0.175	21 000	0.390
Kevlar	x	198 000	x	x	0.903

TABLE II. A compiled list of the nonlinear parameter values for the five materials studied. An x for Kevlar indicates that the value was not computable. In the table, "freq" is the frequency in Hz which indicates the frequency dependence of the parameters.

material	U_c (m/s)	r_1 (Rayls s/m^2)	m_1 (kg s/m^4)	frequency range
30 ppi foam	0.15	375	$-0.145 + 2.87\times10^{-4}$ freq	freq < 500 Hz
100 ppi foam	0.52	2350 − 1.68 freq	−0.455	freq < 500 Hz
0.5 mm beads	0.2	72 900 − 7.31 freq	−10.5 + 0.0058 freq	freq < 800 Hz
1 mm beads	0.3	27 300 − 3.91 freq	−7.65 + 0.0048 freq	freq < 1000 Hz
Kevlar	0.15	16 000	x	

mogeneities caused by placing many samples together in the measurement tube to achieve an infinite sample approximation. Apparently, either the samples were slightly dissimilar, or the finite-amplitude excitation caused a slight sample separation where they came together which produced poor measured results.

Second, due to the natural resonances of a sample, higher particle velocities could be achieved at lower surface pressures than could be obtained for an infinite sample. Higher particle velocities imply enhanced nonlinear viscous effects. Thus, by using finite-length samples and exciting them about their resonant frequencies, we could observe greater nonlinear behavior in the surface admittance. Also, as noted in Ref. 1, a finite-length sample can actually experience a peaking in its sound absorptivity for increasing surface pressure. However, this phenomena will not be further discussed in this paper having been observed and commented on in Ref. 1.

Shown in Fig. 9, is admittance data for a 9.25-cm sample of 100-ppi foam. Since the frequencies of measurement, 900 and 992 Hz, lie outside of the frequency range of Table II, the nonlinear parameters r_1 and m_1 were chosen so that the nonlinear effects were best predicted. As given in the figure caption, an r_1 of 1300 and an m_1 of zero were used. That both r_1 and m_1 should continue to decrease with frequency is consistent with the previous discussion and the supposition that m_1 goes to zero is also supported. However, the most important feature of Fig. 9 is how well the theoretical predictions match the experimental values as compared to that of Fig. 5 of Ref. 1. The only difference in the model used in Ref. 1 and that put forth here is the quadratic nonlinear regime of R and M below the critical velocity U_c. In Ref. 1, the nonlinear correction to R took the form of

$$R(\omega,|U|) = r_0(\omega) + \eta_0|U|/\sqrt{2}. \tag{26}$$

Although similar to Eq. (22a), the important difference is that it predicts a greater nonlinear resistance for particle velocities below U_c. The effect of this is that a greater nonlinear behavior for finite amplitudes with particle velocities below U_c will be predicted using Eq. (26) than if Eq. (22a) were used. This is born out in the figures. In Fig. 4 of Ref. 1, the predicted admittance shows a more gradual change than is seen in the data. For instance, the data show almost no change from its low pressure values up to 130 dB while the theory, based upon Eq. (29), is already predicting a significant departure. On the other hand, with the quadratic regime correction of Eq. (22a), the prediction of the admittance in Fig. 9 now follows the data almost precisely. Thus the admittance data also support the idea that the departure of R from a constant is not linear for low particle velocities. Due to the results of Eq. (20), we have chosen to characterize this departure as a quadratic. Although this departure seems slight, it may be of importance for a more fundamental understanding of the nonlinear process in porous materials.

Figure 10 shows similar admittance data for 30-ppi foam. In this case, one frequency, 610 Hz, is near a resonant frequency while the other, 825 Hz, is near an antiresonant frequency. The effect is as expected: Near a resonant frequency, velocities will be "large," thus nonlinear effects will be most prominent; near an antiresonant frequency, velocities will be "small," thus nonlinear effects will be less evident. Shown in Fig. 11, is admittance data for a 9.1-cm sample of 1-mm beads at 508 Hz. In this plot (unlike the previous two) a nonzero m_1 coefficient was used in the theo-

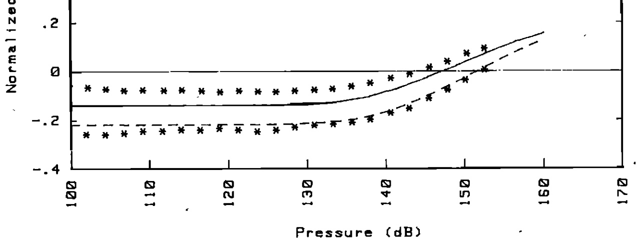

FIG. 9. Normalized surface admittance of a 9.25-cm sample of 100-ppi foam versus surface SPL for two frequencies, 900 and 992 Hz. Pluses are conductance values (g), while the asterisks are susceptance values (b). The solid line is a prediction of 900-Hz data while the dashed line is for the 992-Hz data. Parameters used in prediction are: $\Omega = 0.974$, $a_c = 0.166$ mm, $\sigma_0 = 5470$, $r_1 = 1300$ mks Rayls s/m^2, $m_1 = 0$, and $U_c = 0.52$ m/s. For a plot of how the admittance of this sample varies with frequency, see Fig. 4 of Ref. 1.

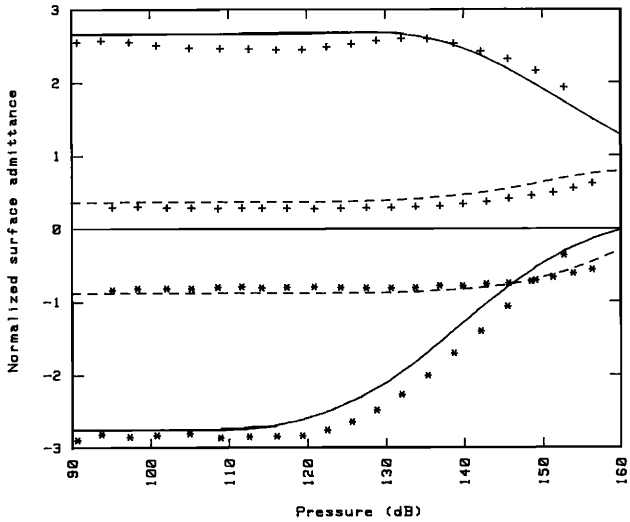

FIG. 10. Normalized surface admittance of a 14.7-cm sample of 30-ppi foam versus surface SPL for two frequencies, 610 and 825 Hz. Pluses are conductance values (g) while the asterisks are susceptance values (b). The solid line is a prediction of 610-Hz data while the dashed line is for the 825-Hz data. 610 Hz is near a sample resonant frequency while 825 is near an antiresonant frequency. Parameters used in prediction are: $\Omega = 0.973$, $a_c = 0.7096$ mm, $\sigma_0 = 295$, $r_1 = 375$ mks Rayls s/m^2, $m_1 = 0$, and $U_c = 0.15$ m/s.

retical prediction. As can be seen in comparing the two theoretical curves for the admittance (solid and dashed), a nonzero m_1 value improves the prediction of the susceptance, while the conductance prediction suffers. However, the net

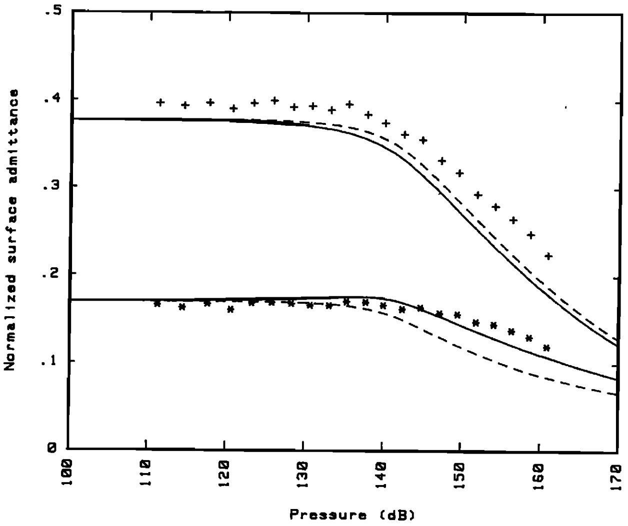

FIG. 11. Normalized surface admittance of a 9.1-cm sample of 1-mm glass beads versus surface SPL for a frequency of 508 Hz. Pluses are conductance values (g) while the asterisks are susceptance values (b). The solid line is a prediction using a m_1 coefficient of -5.12 kg s/m^2, while the dashed line is for a zero m_1 coefficient. Other parameters used in prediction are: $\Omega = 0.390$, $a_c = 0.178$ mm, $\sigma_0 = 19500$, $r_1 = 25400$ mks Rayls s/m^2, and $U_c = 0.30$ m/s.

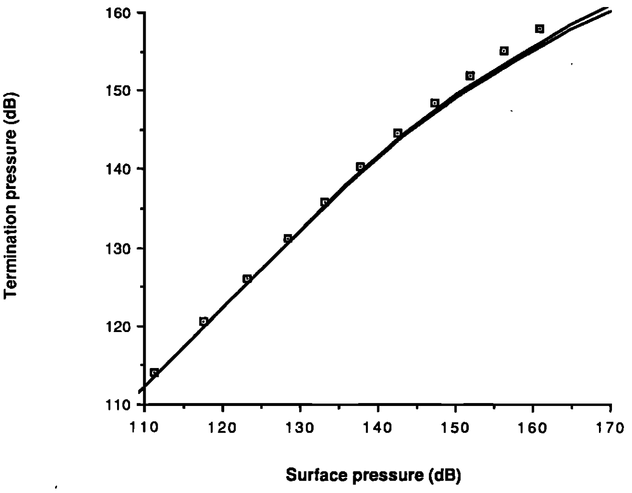

FIG. 12. Termination pressure versus surface pressure of the 1-mm glass bead sample of Fig. 11, again at a frequency of 508 Hz. Parameters are the same as given in Fig. 11. Upper solid line is for $m_1 - 0.0$ while the lower solid line is for $m_1 = -5.12$ kg s/m^2.

effect is taken to be an improvement over a purely resistive nonlinear correction as has been previously used.[9,16]

Finally, data were taken of surface pressure versus termination pressure for a finite-length sample of 1-mm glass beads under the same excitation for which the admittance data were obtained (fixed frequency pure tone). [The termination pressure was obtained by placing a 1/2-in. B&K microphone in the termination wall as shown in Fig. 1 of Ref. 2. Indeed, the admittance data and the pressure data for the 30-ppi foam and 1-mm beads were obtained using the apparatus of Ref. 2, where the length L was reduced to zero (see referenced figure for explanation of variables). The admittance becomes the ratio of U_1 to P_1 and the termination pressure is the pressure of microphone one, P_{m1}.]

Figure 12 shows such pressure data for the sample of Fig. 11. Note that, as was also observed by Kuntz and Blackstock[9] for infinite length samples, the termination pressure falls off from a linear relation with the surface pressure. The presence of a nonlinear mass term is seen to decrease the sound penetrating into the sample as denoted by the lower of the two curves in Fig. 11 (see caption). This can be intuitively understood as follows: Since the resistance R that the fluid experiences increases with increasing fluid velocity, more of the wave energy is reflected back as the intensity is increased. Thus less wave energy will penetrate into the sample causing the pressure to be lower than that predicted by linear theory. Also, since more of the wave is reflected back to the source, the absorption coefficient will correspondingly decrease with increasing intensity. It was calculated that at a surface pressure of 170 dB, the nonzero m_1 value of -5.12 kgs/m^2 will cause the mass term to become zero at the surface. Examining Fig. 11 shows that this condition was never reached for this material.

(For a discussion of the errors observed in the fixed frequency plots, see the discussion at the end of Sec. IV in Ref. 17.)

V. APPROXIMATE NONLINEAR SURFACE PRESSURES

Since the surface pressure can be related to the surface admittance and particle velocity via the formula $P = U/Y$, one can estimate the surface pressure where nonlinearities begin to occur if a criterion is known for the velocity and we have some estimate of the surface admittance. As we have seen, nonlinearities for these materials begin to be noticeable at their critical velocity U_c. Also, for these porous materials, it can be shown that the high frequency limit of their surface admittance Y is $\Omega/\rho_0 c$. If we call the critical pressure where the nonlinearities begin to effect the surface pressure P_{nl}, then, using the two criteria given above, we obtain

$$P_{nl} = U_c \rho_0 c/\Omega. \tag{27}$$

Using Eq. (27) and the values from Tables I and II, we obtain the following results for air:

material	P_{nl}
30-ppi foam	130 dB
100-ppi foam	141 dB
0.5-mm beads	144 dB.

Comparing these values to the materials corresponding nonlinear admittance data of Figs. 9–11, we see that there is a very good agreement with the pressures stated above and the pressures where the admittances begin to show nonlinear behavior. Although Eq. (27) is very simple, it does an excellent job of predicting the pressure at which one would need to become concerned with finite-amplitude effects when dealing with a porous material.

More accurate estimations for a P_{nl}, valid for all frequencies, can be obtained using a better estimate for Y. The admittance Y is of course dependent upon the type of acoustical system under consideration. For a given system, it is usually not too difficult to obtain a fairly accurate estimate of Y using linear theory. What then remains is to find an estimate for the particle velocity U. Here it was convenient to use the critical velocity U_c. However, for most materials, a value for U_c will not be known. Since most of the literature on the nonlinear behavior of porous materials deals with only a linear deviation of the static flow resistance σ from a constant (the parameter by which it varies of course being the particle velocity), an estimate for a P_{nl} would probably have to make use of that. Thus, if $\sigma(u_{dc}) = \sigma_0 + \eta u_{dc}$, one such possible estimate for a U would be when the nonlinear term becomes of the order of magnitude of σ_0. In this case one obtains an estimate for P_{nl} of

$$P_{nl} = \sigma_0/\eta|Y|. \tag{28}$$

However, the accuracy of this prediction has not been tested.

VI. CONCLUDING REMARKS

In this paper, we have endeavored to parametrize the viscous nonlinearities of fluid saturated open cell porous acoustic materials beyond that of a simple linear velocity dependence correction to Darcy's law.[1,9,16] It was observed that the nonlinear coefficients r_1 and m_1 are frequency dependent. Unlike experiments carried out by Zorumski and Parrott on thin perforates,[14] the r_1 coefficient decreases with increasing frequency, while m_1 increases from a negative value to zero with increasing frequency. One new contribution of this work is in putting forth and giving mathematical support for the existence of two nonlinear flow regimes separated by a critical (or transition) velocity U_c. How important accurate knowledge of the critical velocity of a porous material is in practical applications cannot be guessed but it does provide a simple guide for predicting an approximate surface SPL where nonlinear viscous phenomena become important in describing the behavior of porous materials. Such information may also be important to the further understanding of these nonlinear mechanisms. No satisfactory physical explanation for the existence of two nonlinear flow regimes characterized by a transition or critical particle velocity U_c is known to the authors. Nevertheless, the nonlinear characterization of parameters as expressed in Eq. (22) does a very good job of predicting nonlinear surface behavior of a variety of porous materials at high surface sound-pressure levels.

ACKNOWLEDGMENTS

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